2025 AMC 10A Problem 9 - Solution

Review the full statement and step-by-step solution for 2025 AMC 10A Problem 9. Great practice for AMC 10, AMC 12, AIME, and other math contests

2025 AMC 10A Problem 9

Problem:
Let $f(x) = 100x^3 - 300x^2 + 200x$. For how many real numbers a does the graph of $y = f(x-a)$ pass through the point $(1, 25)$?

Answer Choices:

A. 1

B. 2

C. 3

D. 4

E. more than 4

Solution:

We need the graph of $y = f(x-a)$ to pass through $(1, 25)$.

That means

$$f(1-a) = 25.$$

Given

$$f(x) = 100x^3 - 300x^2 + 200x = 100x(x-1)(x-2),$$

we have

$$\begin{aligned} f(1-a) &= 100(1-a)(1-a-1)(1-a-2) \\ &= 100(1-a)(-a)(-1-a) \\ &= 100a(1-a)(1+a) \\ &= 100a(1-a^2) \end{aligned}$$

So

$$100a(1-a^2) = 25 \quad \Longrightarrow \quad 4a(1-a^2) = 1,$$

$$4a - 4a^3 = 1 \quad \Longrightarrow \quad 4a^3 - 4a + 1 = 0.$$

Let $g(a) = 4a^3 - 4a + 1$. Then

$$g'(a) = 12a^2 - 4 = 4(3a^2 - 1),$$

so critical points at

$$a = \pm\frac{1}{\sqrt{3}}.$$

We can evaluate

$$g\left(-\frac{1}{\sqrt{3}}\right) > 0, \qquad g\left(\frac{1}{\sqrt{3}}\right) < 0.$$

Since $g(a) \to -\infty$ as $a \to -\infty$ and $g(a) \to +\infty$ as $a \to +\infty$, the cubic crosses the a-axis once in each of the intervals

$$\left(-\infty, -\frac{1}{\sqrt{3}}\right), \quad \left(-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right), \quad \left(\frac{1}{\sqrt{3}}, \infty\right),$$

giving 3 distinct real roots.

Therefore, there are

$$\boxed{\textbf{(C) } 3}$$

real numbers a such that the graph of $y = f(x-a)$ passes through $(1, 25)$.